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MLA0201-Fundamentals Of Machine Learning

Comparative Analysis

Question 1

Bayesian Networks vs. Markov Random Fields for Disease Diagnosis

Healthcare diagnosis systems must represent how diseases give rise to observable symptoms, and how prior medical history influences the likelihood of a condition. Two major classes of probabilistic graphical models are commonly used for this purpose: Bayesian Networks (directed graphical models) and Markov Random Fields (undirected graphical models). Both encode joint probability distributions over a set of random variables, but they differ fundamentally in structure, independence assumptions, and inference behaviour, which affects their suitability for a disease-symptom modeling task.

Comparative Analysis

Criterion	Bayesian Network	Markov Random Field
Graph Structure	Uses a Directed Acyclic Graph (DAG) where edges point from cause to effect, e.g. Disease -> Symptom. The direction naturally mirrors the causal relationship between a disease and its clinical manifestations.	Uses an undirected graph where edges represent mutual correlation rather than causation, e.g. Symptom - Symptom or Disease - Disease. There is no explicit notion of cause and effect.
Conditional Independence	Encoded through the concept of d-separation. A node is independent of its non-descendants given its parents, which fits naturally with the idea that symptoms are conditionally independent given the disease that causes them.	Encoded through the Markov blanket and the global Markov property: a node is independent of all other nodes given its immediate neighbours. This is useful for modeling symmetric, mutually reinforcing

Criterion	Bayesian Network	Markov Random Field
		relationships such as co-occurring symptoms.
Probability Representation	Joint distribution factorizes as a product of conditional probability distributions, $P(X) = \text{product of } P(X_i \text{ given Parents}(X_i))$. Each conditional probability table has a direct clinical interpretation, such as $P(\text{Fever given Malaria})$.	Joint distribution factorizes as a product of potential functions over cliques, $P(X) = (1/Z)$ times the product of clique potentials. These potentials are not probabilities themselves and often lack an intuitive clinical meaning.
Inference Process	Supports efficient exact inference in sparse networks using algorithms such as variable elimination and belief propagation on tree-structured or moderately connected graphs. Diagnostic (symptom to disease) and predictive (disease to symptom) reasoning are both natural.	Inference generally requires computing the partition function Z , which is intractable for large or densely connected graphs. Approximate methods such as loopy belief propagation, Gibbs sampling, or variational inference are commonly needed.
Uncertainty Handling	Naturally represents uncertainty through conditional probability tables and supports probabilistic queries such as the likelihood of a disease given observed symptoms, including prior medical history as evidence nodes.	Represents uncertainty through potential functions, which is effective for capturing soft, symmetric constraints among symptoms but less effective for expressing directional causal uncertainty such as how a pre-existing condition influences a new disease.
Computational Complexity	Complexity depends on network sparsity; for typical disease-symptom networks with a moderate number of parents per node, exact inference is often tractable. Learning parameters	Computing the normalizing constant Z is NP-hard in general, making both learning and inference computationally expensive, especially as the

Criterion	Bayesian Network	Markov Random Field
	from data (frequencies) is straightforward using maximum likelihood estimation.	number of interacting symptoms and diseases grows.
Real-World Applicability	Widely used in clinical decision support systems (for example the classic QMR-DT and Pathfinder systems), since it mirrors how physicians reason from diseases to symptoms and can seamlessly incorporate patient history as prior probabilities.	More applicable to problems such as medical image segmentation or spatial labeling tasks, where neighbouring pixels or regions have symmetric dependencies, rather than to explicit disease causation reasoning.

Justification and Recommendation

For the healthcare diagnosis application described, a Bayesian Network is the more appropriate choice. Diseases and symptoms have a naturally directed, causal relationship: a disease increases the probability of certain symptoms appearing, not the other way around. Bayesian Networks allow this causal structure to be encoded directly in the graph, make it possible to incorporate a patient's medical history as prior evidence, and support efficient diagnostic inference through algorithms such as belief propagation. The conditional probability tables are also interpretable by clinicians, which builds trust in the system's recommendations. Markov Random Fields, while powerful for symmetric spatial or relational data, do not capture the directionality of disease causation as naturally and typically demand more expensive inference procedures. Hence, Bayesian Networks are recommended as the primary modeling framework for this intelligent disease diagnosis system.

Question 2

Hidden Markov Models vs. Conditional Random Fields for Driving Activity Recognition

Recognizing driving activities such as accelerating, braking, turning, or lane changing from continuous sensor streams is a sequence labeling problem, where a label must be assigned to each time step based on the observed sensor readings and the surrounding temporal context. Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs) are the two classical approaches used for this class of problem, and they differ substantially in how they model the relationship between hidden states and observations.

Comparative Analysis

Criterion	Hidden Markov Model	Conditional Random Field
State Representation	Models a sequence of hidden states (driving activities) using first-order Markov assumption, where the current state depends only on the previous state. Generative model: it models the joint probability $P(\text{states, observations})$.	Models the conditional probability of the label sequence given the observation sequence, $P(\text{states given observations})$, directly. Discriminative model: it does not attempt to model how observations are generated.
Observation Modeling	Assumes observations at each time step depend only on the current hidden state and are conditionally independent of each other given that state, which is often unrealistic for correlated multi-sensor data such as GPS, accelerometer, and steering angle readings together.	Allows arbitrary, overlapping features of the observation sequence (past, present, and future sensor readings) to influence the label at each time step, without assuming independence between observations.
Learning Approach	Learned using Maximum Likelihood Estimation via the Baum-Welch (Expectation-Maximization) algorithm, estimating transition and emission probabilities from labeled or partially labeled sequences.	Learned by directly optimizing the conditional log-likelihood of label sequences given observations, typically via gradient-based methods, which tends to produce better calibrated boundaries for the actual prediction task.

Criterion	Hidden Markov Model	Conditional Random Field
Prediction Accuracy	Performs reasonably well for simple, well-separated activities but accuracy degrades when sensor features are highly correlated or when the true dependency structure violates the independence assumption.	Generally achieves higher labeling accuracy for complex activity recognition tasks because it can use rich, overlapping, and long-range features directly relevant to distinguishing similar driving manoeuvres.
Handling of Contextual Features	Limited ability to incorporate rich contextual features; extending an HMM to use multiple correlated features typically requires strong independence assumptions that reduce modeling accuracy.	Naturally incorporates rich contextual and overlapping features such as speed change over a window, road curvature, and nearby vehicle proximity, all combined into a single feature-based framework.
Computational Efficiency	Computationally efficient; training and inference (via the Viterbi algorithm) scale well and are suitable for real-time or embedded automotive systems with limited computational resources.	More computationally intensive during training because it requires computing a global normalization term over all possible label sequences, though inference at run-time (Viterbi-style decoding) remains comparable in cost to HMMs.
Scalability	Scales well to large volumes of streaming sensor data and long driving sessions due to lower training cost, but scalability to a large number of complex features is limited.	Scales well with respect to the number and richness of features, but training time and memory usage grow with sequence length and feature-set size, requiring more computing power for large-scale deployment.

Recommendation and Justification

For an autonomous vehicle activity recognition system, a Conditional Random Field is the more suitable model overall, provided sufficient computational resources are available for training. CRFs can combine multiple correlated sensor streams (accelerometer, GPS, steering angle, braking pressure) are controllable.

Question 3

Directed vs. Undirected Graphical Models for Financial Risk Prediction

A financial risk prediction system must capture dependencies among many variables, such as income level, credit history, transaction patterns, existing debt, and macroeconomic indicators, in order to estimate the probability of default or fraud. Directed Graphical Models (Bayesian Networks) and Undirected Graphical Models (Markov Random Fields) offer two different ways of representing these dependencies, each with distinct implications for financial modeling.

Comparative Analysis

Criterion	Bayesian Network (Directed)	Markov Random Field (Undirected)
Graph Representation	Directed Acyclic Graph where edges express a hypothesized causal or influence direction, for example Income -> Credit Score -> Default Risk, allowing analysts to encode domain knowledge about how financial factors influence one another.	Undirected graph where edges represent symmetric statistical association, for example between correlated spending categories, without asserting which variable influences which.
Dependency Modeling	Models dependencies through conditional probability distributions of each variable given its parents, which is well suited to expressing structured expert knowledge about how financial indicators causally relate to risk.	Models dependencies through clique potential functions, which is well suited to expressing mutual, non-causal correlations among transactions, such as clusters of related spending behaviours occurring together.
Inference Techniques	Supports exact inference via variable elimination or junction tree algorithms for moderately sized networks, and approximate inference (sampling, belief propagation) for larger networks; allows both forward risk prediction and backward diagnostic queries.	Requires computing the partition function for exact inference, which is generally intractable at scale; approximate inference through Markov Chain Monte Carlo methods or loopy belief propagation is typically used instead.

Criterion	Bayesian Network (Directed)	Markov Random Field (Undirected)
Learning Algorithms	Structure can be learned using score-based (e.g. BIC) or constraint-based methods, and parameters estimated via maximum likelihood or Bayesian estimation from historical financial records, often incorporating expert-defined structure.	Parameter learning typically requires iterative optimization such as gradient ascent on the pseudo-likelihood or contrastive divergence, since direct maximum likelihood estimation is hindered by the intractable normalization constant.
Generalization Capability	Generalizes well when the underlying causal structure between financial variables is reasonably well understood and stable over time, and is robust to missing data through probabilistic imputation.	Generalizes well to capturing complex, symmetric correlation patterns among many transaction features, such as fraud rings or correlated spending clusters, even without a clear causal explanation.
Computational Requirements	Computationally efficient for sparse, moderately sized networks typical of curated financial risk factors; interpretable structure reduces the effort needed for regulatory model validation.	Higher computational cost due to the intractable partition function, especially as the number of correlated transaction features grows, making large-scale deployment more resource-intensive.
Suitability for Financial Risk Prediction	Well suited to producing interpretable, auditable risk scores that regulators and analysts can trace back through cause-and-effect reasoning, an important requirement in financial services.	Better suited to auxiliary tasks such as detecting anomalous correlation patterns or fraud clusters, where relationships are mutual rather than causal.

Conclusion and Recommendation

For the financial risk prediction system described, a Bayesian Network is the more effective overall choice. Financial risk assessment benefits strongly from causal interpretability, since regulators.